

Sparse data limit what a mechanistic corrosion model can predict

Identifiability of a point defect model for oxide growth on AISI 347

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The question

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When we fit a mechanistic corrosion model to a handful of measurements and extrapolate it across a service lifetime, **what have the data actually determined?**

Not: does the model fit. It does.

But: would a very different parameter set have fitted just as well, and predicted something else entirely?

A model that fits is not the same as a model that is *identified*. If the data cannot separate two parameter sets, the fit is honest and the extrapolation is still meaningless.

The problem

The problem

Mechanistic corrosion models are routinely

- written with four to six adjustable parameters,
- fitted to a handful of measurements,
- and then extrapolated across a service lifetime.

What is almost never tested

Whether the data determine those parameters at all.

The distinction is inert while the model is used inside its calibration window, and decisive the moment it is used outside one, which is the only reason to build a mechanistic model in the first place.

The gap

The gap in the literature

Point defect model parameter estimation goes back four decades.

What is done

- PDM fitted to impedance spectra, or to depth profiles
- four to six parameters against three to six data points
- confidence intervals and correlations *sometimes* reported

What is missing

- no profile-likelihood characterisation
- no bootstrap identifiability analysis
- no separation of prior leverage from data leverage

To our knowledge this is the first formal identifiability characterisation of a point defect model fit. The tools are mature in systems biology and in battery modelling; they have not been brought here.

And for AISI 347 in this environment, prior work stops at empirical per-element power laws: no shared physical parameter set, no assessment of how far the fits extrapolate.

The system and the data

The system

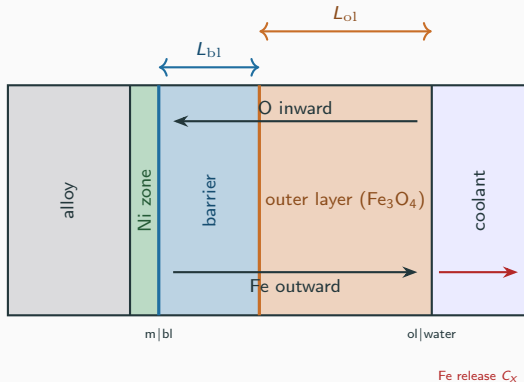
Nb-stabilised austenitic stainless steel

X6CrNiNb18-10 (AISI 347)

Simulated boiling-water-reactor hydrothermal water

- 240 °C, 7 MPa
- 0.4 ppm dissolved O₂
- exposures 72, 168, 480 h

A **duplex** oxide forms: a thin Cr-rich barrier layer that controls the kinetics, and a thicker, far more variable Fe-rich outer layer.



The barrier layer sets the kinetics; the outer layer is where the information runs out.

All of the calibration data

t (h)	Cr barrier (nm)	Fe outer (nm)	scans
72	37.3 ± 7.7	143.3 ± 111.5	3
168	98.3 ± 21.7	225.5 ± 142.7	4
480	134.3 ± 5.4	240.3 ± 172.5	3

Six numbers

Three Cr, three Fe. Five fitted parameters.

Extrapolated 180-fold

Last observation 480 h $\rightarrow$ target 87 600 h.

The Fe standard deviations are 50 to 70% of the means. Whatever the model says about the outer layer, the data barely constrain it.

The model

The reduced point defect model

Barrier layer, from point defect kinetics with a field that relaxes as the film thickens:

$$\frac{dL_{bl}}{dt} = A_{bl} e^{b_3 L_{bl}} - C_{bl}, \quad L_{bl}(0) = L_0.$$

Outer layer, fed by cations crossing the barrier and losing mass to the coolant:

$$\frac{dL_{ol}}{dt} = \text{PBR}_{\text{eff}} \frac{dL_{bl}}{dt} + C_x, \quad L_{ol}(0) = L_{ol,0}.$$

One-way coupling, and why it matters

L_{bl} never reads PBR_{eff} , C_x or $L_{ol,0}$. Every barrier-layer result, including the ten-year prediction, is structurally immune to everything the outer layer does. This is what lets us report an outer-layer non-result without contaminating the headline number.

Reduced from a fourteen-reaction network; only four reactions displace a boundary, and the constant-volume closure eliminates the cation fluxes.

Closed form, and what we verify against it

With $C_{bl} = 0$ the barrier equation integrates exactly:

$$L_{bl}(t) = L_0 - \frac{1}{b_3} \ln\left(1 - b_3 A_{bl} e^{b_3 L_0 t}\right).$$

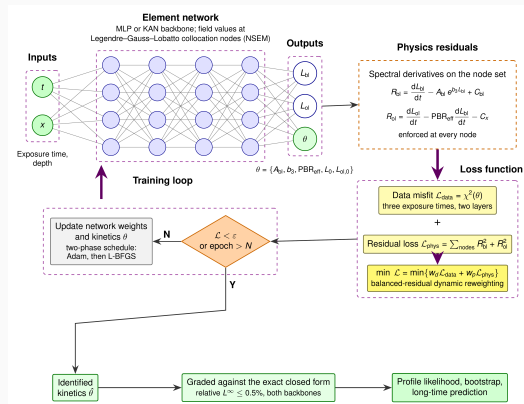
Why this matters more than it looks

A closed form means the inversion can be **graded against an exact reference** rather than trusted. Most physics-informed inverse problems have no such reference. This one does, and that is what makes the failure mode later in this talk visible at all.

Closed form against stiff numerical integration: agreement to better than 6×10^{-11} nm. Any misfit we report is a statement about the data and the model, never about the solver.

Our approach

Our approach



A **differentiable** inversion: exact parameter gradients through a vectorised solver. Kinetics are recovered by hard-constrained inversion, so the governing equations hold by construction, and checked against a classical Levenberg–Marquardt fit.

Identifiability, built in rather than bolted on

Three tests, each capable of falsifying the fit:

1. **Profile likelihood** across each parameter, with the others re-optimised.
2. **Parametric bootstrap**, 1000 members, resampling at the measured scatter.
3. **Prior relaxation**, to check a conclusion is not an artefact of a prior.

What makes this affordable

Differentiability, not the neural representation. Exact gradients are what buy a thousand refits, sixty-one-point profile scans and a dense operating-envelope grid. Closed-form models with automatic differentiation share that property.

And one comparison throughout: the same data fitted with a purely empirical power law, then both extrapolated.

Our code

An installable package

htw-pdm, Apache 2.0

github.com/Feugmo-Group/x6crninb18-pdm

- classical fits, mass balance and spatial solves need only NumPy, SciPy, Matplotlib
- the neural inversions add the NSEM solver
- **71 tests** pin the reported numbers

A committed artefact per number

quantity	artefact
parameters	table1_*.csv
model ladder	table2_*.csv
F-1 exhibit	table4_*.csv
bootstrap, AIC	referee_stats.json

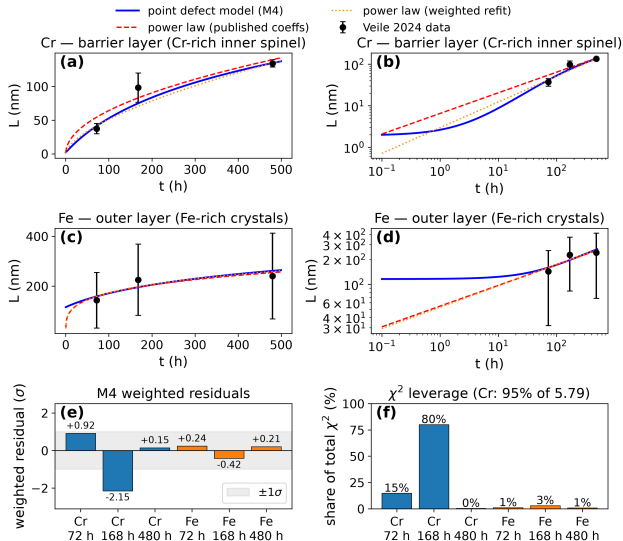
A generated document maps every figure and table to the exact command that produces it.

The rule we work to

No number appears in the paper or on these slides unless a committed artefact produces it. Retyping from memory is how the two drift.

Results

Result 1: the model fits



Accepted model **M4**:

$$A_{bl} = 0.7269 \text{ nm/h}$$

$$b_3 = -0.012487 \text{ nm}^{-1}$$

$$\text{PBR}_{\text{eff}} = 1.096$$

$$L_0 = 1.924 \text{ nm}$$

$$L_{ol,0} = 115.6 \text{ nm}$$

$$\chi^2 = 5.79, \quad \text{dof} = 1$$

Refit under the same weighted objective, the empirical power law reaches $\chi^2 = 7.84$ on 2 dof. Comparable, not better.

Result 2: but the fit is one measurement

Weighted residual decomposition at M4:

	72 h	168 h	480 h
Cr (barrier)	+0.92	− 2.15	+0.15
Fe (outer)	+0.24	−0.42	+0.21

95%

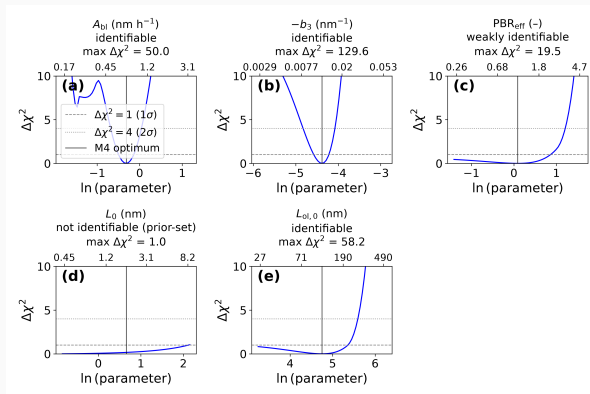
of χ^2 sits on the three chromium points.

80%

sits on the single 168 h chromium point.

The signs run +, −, +: a systematic curvature miss, not scatter. And curvature is exactly what governs the extrapolation.

Result 3: one parameter is not determined at all

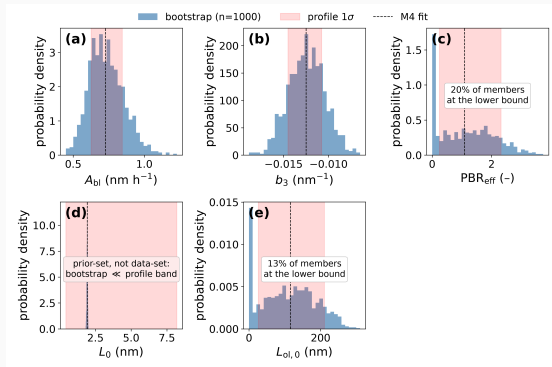


param	max $\Delta\chi^2$	verdict
b_3	129.6	identifiable
$L_{0l,0}$	58.2	identifiable
A_{bl}	50.0	identifiable
PBR_{eff}	19.5	weakly id.
L_0	1.0	unidentifiable

L_0 's profile is **flat**: the data cannot move it. Its apparent bootstrap precision (1.92 ± 0.04 nm) is the prior speaking, not the measurements.

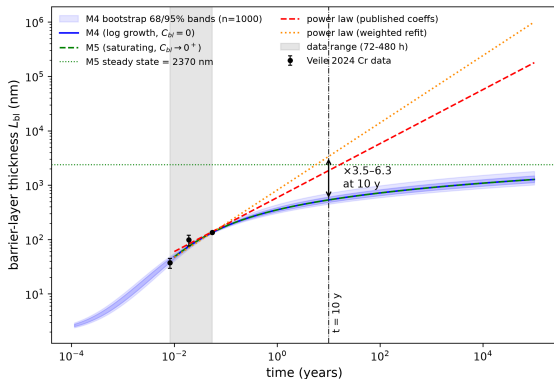
PBR_{eff} spans $[0.25, 2.32]$ at 1σ .

Result 4: profile and bootstrap disagree, and that is the finding



1000 members, zero failures. Two annotated features are results, not plotting artefacts: the **spike at zero** for PBR_{eff} and $L_{ol,0}$ is what weak identifiability looks like under resampling; the **narrow histogram** for L_0 is a prior, not a constraint.

Result 5: the key result



Same data, divergent futures

Ten-year barrier thickness

mechanistic 535 nm

power law 1853 to 3375 nm

a factor of **3.5 to 6.3**

Bootstrap 95% interval on the mechanistic value: [447, 704] nm, widening to roughly [340, 1040] nm under the maximal error rescaling.

Both power-law variants lie outside that band under every error treatment we apply.

What these data determine is the separation, not the number it brackets.

Why you cannot choose between them on the data

	χ^2/dof	AIC	BIC
M4 (mechanistic)	5.79/1	15.79	14.75
power law (refit)	7.84/2	15.84	15.01

$$\Delta\text{AIC} = 0.05$$

On $n = 6$ points the two models are **statistically indistinguishable**, and they disagree six-fold about the thing we actually want to know.

This is the whole argument in one line. Model selection on the calibration window carries essentially no information about the extrapolation.

So: design the experiment that settles it

Exposure needed before the two forms separate:

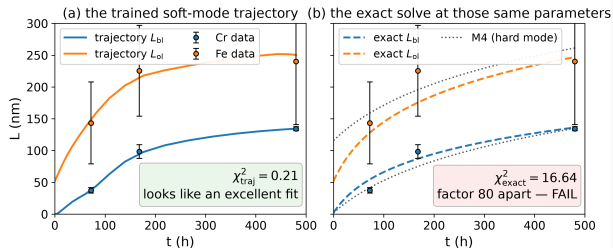
error budget	power law	2σ	3σ
future scatter only	refit	1437 h	2020 h
future scatter only	published	2546 h	4014 h
full budget	refit	1928 h	5493 h
full budget	published	2988 h	5129 h

Recommendation

A single designed exposure of roughly **3000** h discriminates the mechanistic and empirical kinetics at 2σ against the *full* error budget: future scatter, mechanistic parametric uncertainty and power-law fit uncertainty together.

3σ needs 5000 to 6000 h and, because the power law's own uncertainty grows with its prediction, is available only inside a window that closes near 34 000 h.

Result 6: a failure mode the exact reference exposes



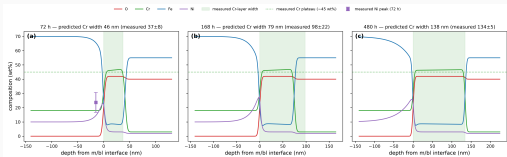
Joint field-and-parameter training settles where the **data loss is satisfied and the governing equation is not.**

mode	χ^2_{traj}	χ^2_{exact}
hard	5.79	5.79
soft, λ_{phys}	0.21	16.64
soft, λ_{data}	0.67	15.57

The trajectory *looks* like a better fit than the accepted model. The parameters it reports are wrong.

Remedy: re-solve the recovered parameters through an independent physics-exact forward solve and report that χ^2 beside the training one. We propose this alongside any physics-informed inverse result.

Result 7: the spatial extension predicts composition for free



Extended spatially with **no additional free parameters**: every kinetic parameter frozen at its zero-dimensional value.

- Cr plateau 46.7 wt% predicted against ~ 45 measured
- layer widths matched to +22%, -20%, +3%

A structural argument, confirmed over four orders of magnitude in diffusivity, shows quasi-steady composition data **cannot** constrain the defect diffusivity however much is taken.

Takeaway

The takeaway

1. A reduced point defect model fits published depth-profile data for AISI 347 in BWR hydrothermal water, but **one of five parameters is undetermined** and a second only weakly determined.
2. **95% of the fit statistic rests on three points and 80% on one**, with a systematic curvature miss. Single-number goodness of fit conceals this.
3. Mechanistic and empirical kinetics are **indistinguishable on the calibration data** ($\Delta AIC = 0.05$) yet differ **3.5 to 6.3 fold** at ten years.
4. What the data determine is **the separation**, not the 535 nm it brackets.
5. **A designed exposure near 3000 h would settle it.**

The transferable point

Report identifiability alongside goodness of fit, or the extrapolation is not evidence. Nothing here is specific to this alloy or to the point defect model.

Limitations

What this study cannot claim

- **Six data points.** No amount of further computation raises the information content. We add no new experiment.
- **The fit is formally rejected** at one degree of freedom: $\chi^2 = 5.79$ gives $p = 0.016$, and the error scale needed to make it acceptable is $s = 2.4$.
- **Part of the data is digitised** from a published figure.
- **The [447, 704] nm interval understates the truth.** It is conditional on the model form, and the power-law comparison shows model-form uncertainty is roughly an order of magnitude larger.
- **One temperature, one oxygen level, unirradiated, static.** The operating-envelope map is a forward projection, not validation.
- **The neural solver was not needed.** Classical methods reproduce every result here, and in the spatial case more accurately. That is the point of a gradeable test case, not a defect of it.

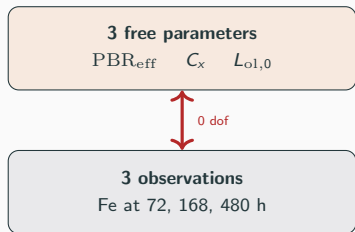
Questions

Backup: the model ladder

model	extra	n_{free}	χ^2	A_{bl}	PBR_{eff}	$L_{\text{ol},0}$
M1 core	—	4	7.15	0.7516	2.433	0
M2	$+C_{\text{bl}}$	5	7.15	0.7516	2.433	0
M3	$+C_{\text{bl}} + C_x$	6	4.11	955.8	4.873	0
M4 (accepted)	$+L_{\text{ol},0}$	5	5.79	0.7269	1.096	115.6
M5	$+L_{\text{ol},0} + C_{\text{bl}}$	6	5.79	0.7269	1.096	115.6
power law (published)	k, n per layer	0	20.10	—	—	—
power law (refit)	k, n per layer	4	7.84	—	—	—

M3 reaches a lower χ^2 by running onto a degenerate ridge ($A_{\text{bl}} \approx C_{\text{bl}} \approx 956$, $b_3 \rightarrow 0$), predicting 155 nm at ten years against M4's 535. It is rejected on structure, not on fit. M5 recovers M4 exactly, confirming C_{bl} is redundant.

Backup: the outer layer is exactly determined, so it says nothing



exactly determined
 $\Rightarrow$ any admissible PBR_{eff} is reachable

PBR_{eff}	$\Delta\chi^2$	C_x (nm/h)
1.096 (fit, $C_x=0$)	—	0
2.053 (mass balance)	-0.120	-0.241
2.156 (with Ni routing)	-0.131	-0.264
2.433 (no precipitate)	-0.157	-0.327

The apparent factor-1.9 discrepancy between the fitted PBR_{eff} and the stoichiometric mass balance is **not a discrepancy**. The stoichiometric value costs nothing; χ^2 even improves.

The data cannot separate iron *production* from iron *loss to the coolant*.

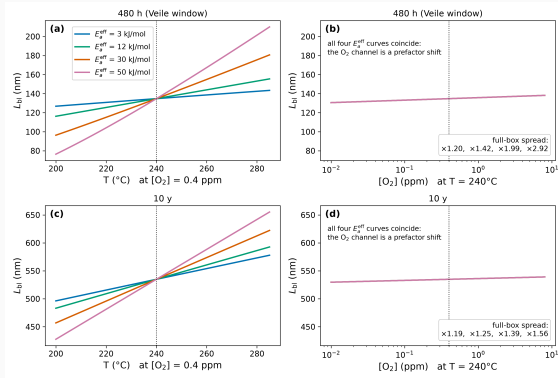
Backup: a falsifiable by-product

Imposing the stoichiometric ratio forces a specific outer-layer loss rate, and therefore a specific iron release into the coolant:

PBR_{eff} imposed	C_x (nm/h)	Fe release ($\mu\text{g dm}^{-2} \text{h}^{-1}$)
1.096	−0.025	0.94
2.156	−0.264	9.94
2.433	−0.327	12.29

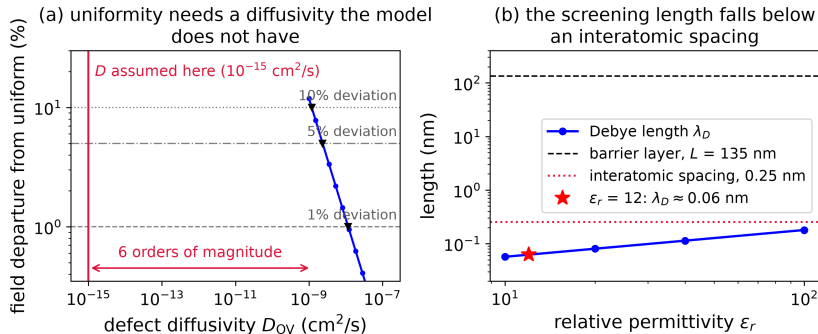
A coolant-chemistry measurement discriminates these by an order of magnitude. The non-result therefore comes with an experiment that would convert it into a result.

Backup: operating envelope



Across 200–285 °C at 7 MPa and 0.01–8 ppm O_2 , the ten-year prediction varies by a factor of 1.19 to 1.56 depending on the assumed activation energy. Small next to the 3.5 to 6.3 that the choice of model form produces.

Backup: does the constant-field closure hold?



Solving Poisson's equation together with the defect transport, instead of imposing the field, shows the PDM's constant-field closure is not supported at the defect concentrations the model itself implies: the implied Debye length is about 0.06 nm, several times smaller than an interatomic spacing. This leaves the fitted kinetics untouched, since the identification is of b_3 and not of the field.

Backup: reproducing every number

All quantities on these slides come from committed artefacts:

slide content	artefact
M4 parameters, profiles	<code>outputs/paper/table1_parameters.csv</code>
model ladder	<code>outputs/paper/table2_model_ladder.csv</code>
F-1 exhibit	<code>outputs/paper/table4_f1_exhibit.csv</code>
envelope factors	<code>outputs/paper/table8_envelope_factors.csv</code>
bootstrap, AIC, power law	<code>outputs/paper/referee_stats.json</code>
PBR_{eff} closure, Fe release	<code>outputs/paper/table10_pbr_closure.csv</code>

The command behind each artefact is listed in `docs/REPRODUCE.md`.